The symbols 1°, $1'$ and $1''$ are used to denote a degree, a minute and a second respectively.

(ii) **Centesimal system**

In the **Centesimal system** , the right angle is divided into 100 equal parts, called **Grades**; each grade is subdivided into 100 Minutes, and each minute is subdivided into 100 Seconds. The symbol 1^g is used to denote a grade.

(iii) **Circular system**

In the **circular system** , the radian measure of an angle is introduced using arc lengths in a circle of radius r. Circular system is used in all branches of Mathematics and in other applications in Science. The symbol 1^c is used to denote 1 radian measure.

3.2.3 Degree Measure

The degree is a unit of measurement of angles and is represented by the symbol $^\circ$. In degrees, we split up one complete rotation into 360 equal parts and each part is one degree, denoted by 1°. Thus, 1° is 1/360 of one complete rotation. To measure a fraction of an angle and also for accuracy of measurement of angles, **minutes** and **seconds** are introduced. One minute $(1')$ corresponds to 1/60 of a degree and in turn a second $(1'')$ corresponds to 1/60 of a minute (or) 1/3600 of a degree.

We shall classify a pair of angles in the following way for better understanding and usages.

(i) Two angles that have the exact same measure are called **congruent angles**.
(ii) Two angles that have their measures adding to 90° are called **complementary angles**.
(iii) Two angles that have their measures adding to 180° are called **supplementary angles**.
(iv) Two angles between 0° and 360° are **conjugate** if their sum equals 360°.

(i) The concept of degrees, minutes and seconds, is analogous to the system of time measurement where we think of a degree representing one hour.

(ii) Observe that

$$\begin{aligned} 59.0854^\circ &= 59^\circ + 0.0854^\circ \\ 0.0854^\circ &= .0854^\circ \times \frac{60'}{1^\circ} = 5.124' \\ 5.124' &= 5' + 0.124' \\ 0.124' &= 0.124' \times \frac{60''}{1'} = 7.44'' \\ \text{Thus, } 59.0854^\circ &= 59^\circ 5' 7.44'' \end{aligned}$$

(iii) Also notice that

$$\begin{aligned} 34^\circ 51' 35'' &= 34.8597^\circ \\ \text{and } 90^\circ - 36^\circ 18' 47'' &= 53^\circ 41' 13'' \end{aligned}$$

3.2.4 Angles in Standard Position

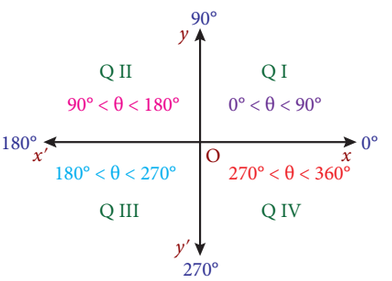


Figure 3.2

An angle is said to be in **standard position** if its vertex is at the origin and its initial side is along the positive x-axis. An angle is said to be in the first quadrant, if in the standard position, its terminal side falls in the first quadrant. Similarly, we can define for the other three quadrants. Angles in standard position having their terminal sides along the x-axis or y-axis are called **quadrantal angles**. Thus, $0^\circ, 90^\circ, 180^\circ, 270^\circ$ and 360° are quadrantal angles.

The degree measurement of a quadrantal angle is a multiple of 90°.